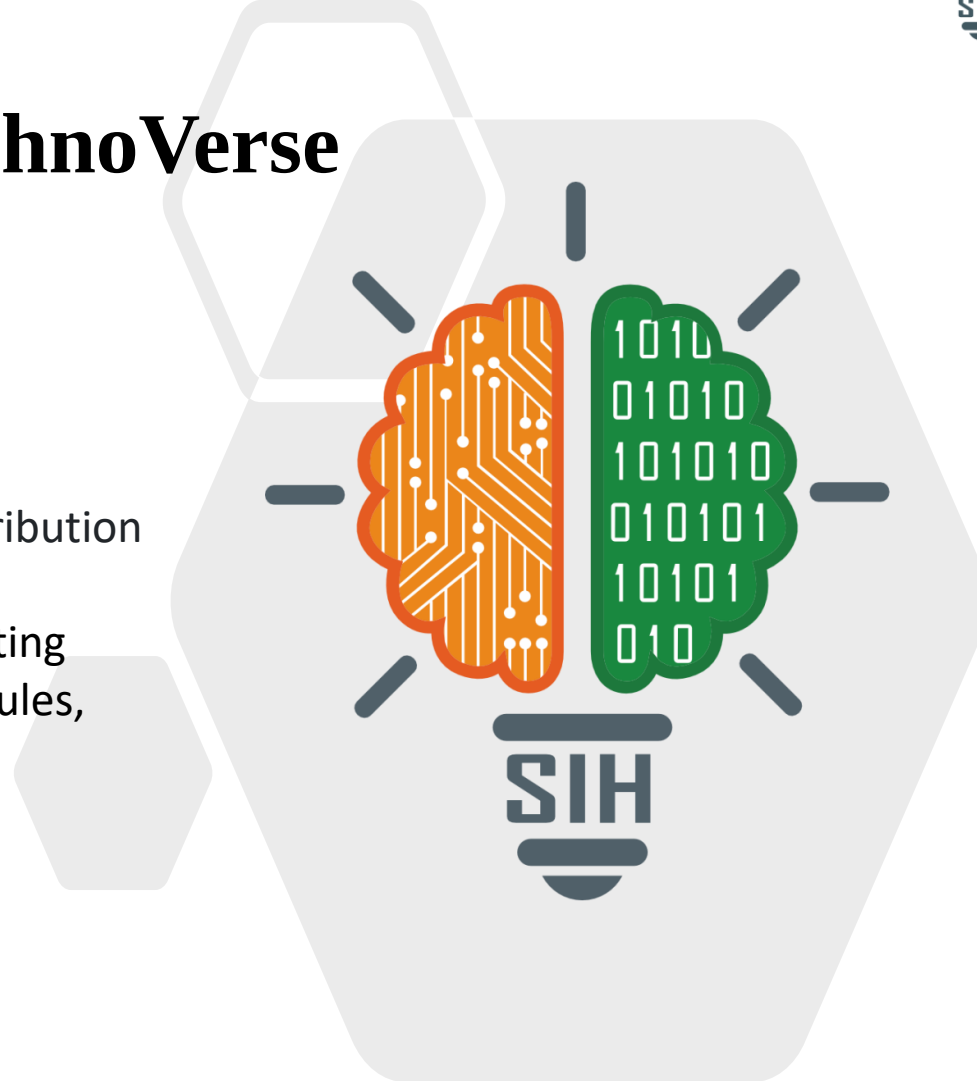


SMART INDIA HACKATHON 2026



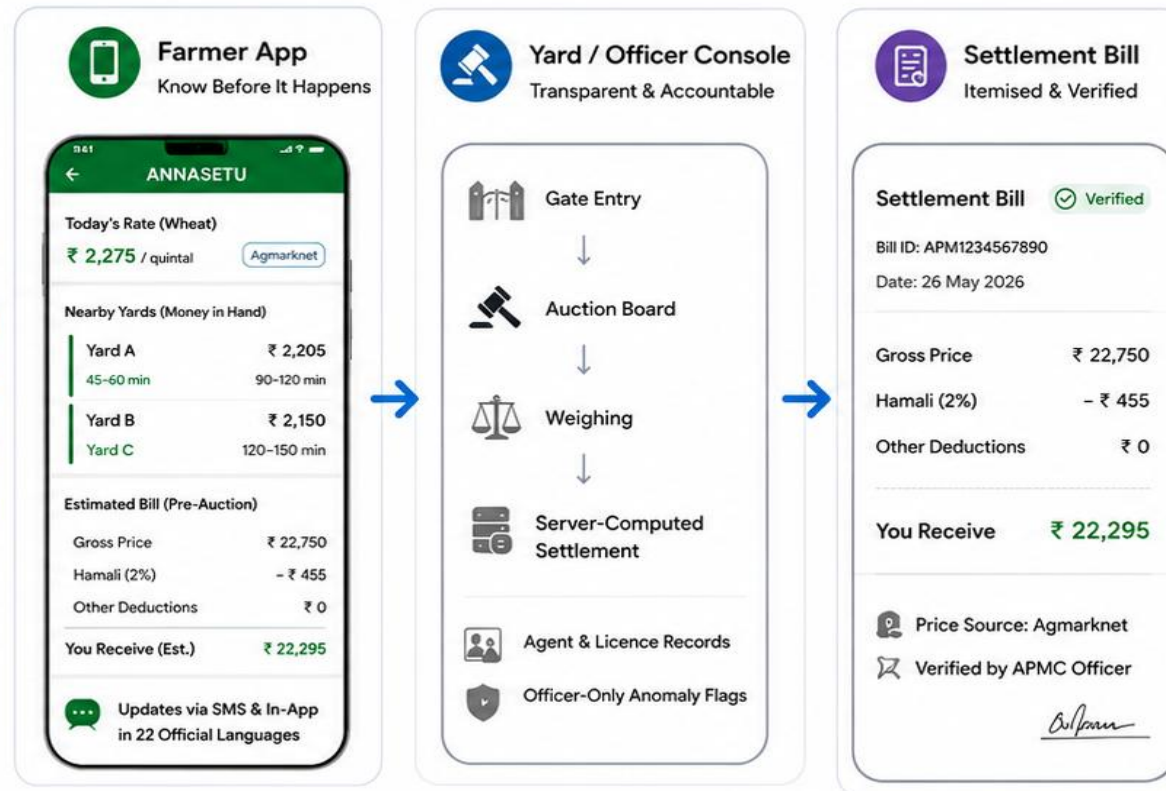
Team: TechnoVerse

- **Problem Statement ID:** SIH26032
- **Theme:** Smart Automation
- **Category:** Software
- **Org.:** Ministry of Consumer Affairs, Food & Public Distribution
- **Problem Statement Title:** Farmers often face long waiting times, lack of information regarding procurement schedules, and uncertainty about procurement status.
- **Team Name:** TechnoVerse
- **Team ID:** TBA (*Team registration not yet open*)



ANNASETU

❖ Proposed Solution



🎯 How it addresses the problem

Replaces blind waiting with a live wait-range, and uncertainty about procurement status with real-time auction & settlement updates on the farmer's phone.

💎 Uniqueness

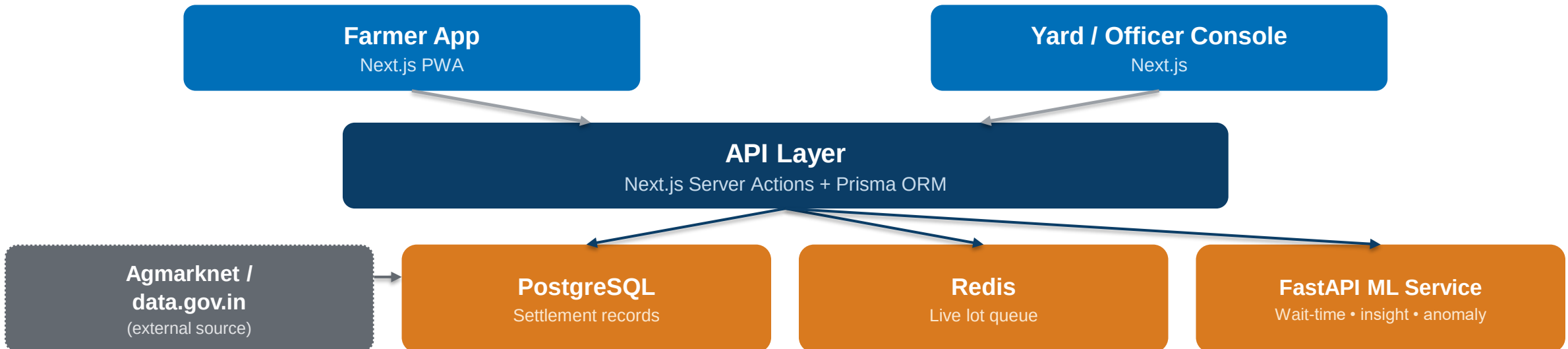
Settlement math verified with an APMC officer (hamali is the only farmer-side deduction); every price is labelled with its provenance; the pre-auction estimate and the issued bill run through the same function, verified byte-identical.

TECH STACK & SYSTEM ARCHITECTURE

TECH STACK



SYSTEM ARCHITECTURE



Live prototype: annasetu-sih.vercel.app — pre-auction estimate and issued bill run through the same function, verified byte-identical

ISOMETRIC WORKFLOW ARCHITECTURE FOR ANNASETU

Clients & Interfaces

- **Farmer app:** Displays rate, value, wait-range.
- **Internal Consoles:** Officer and Agent tools for yard operations (e.g., Gate, Weighing).
- **Notification Service:** Sends SMS & in-app alerts in 22 languages.

Integration & Sync Layer

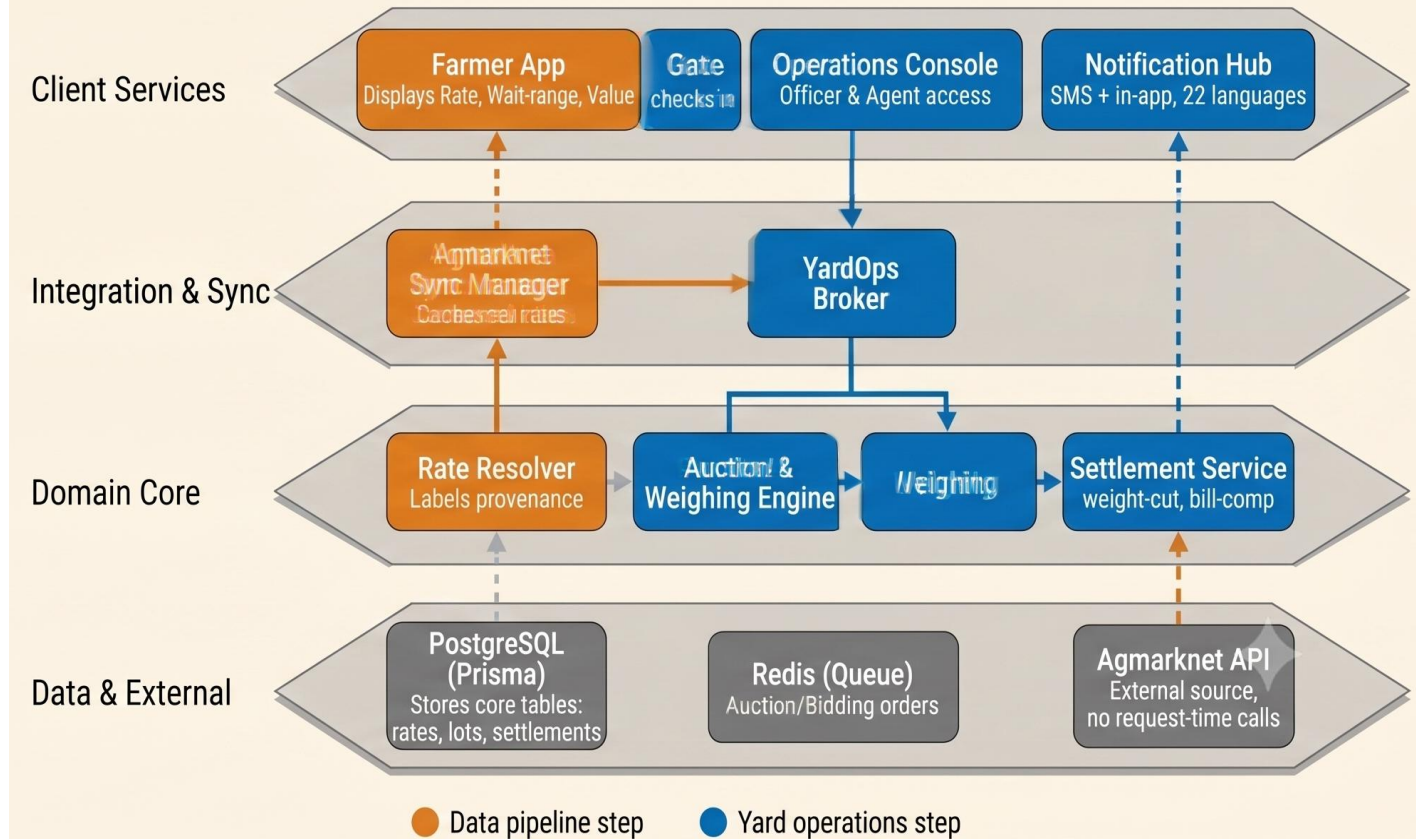
- **Sync Manager:** Caches real Agmarknet rates.
- **YardOps Middleware:** Brokers data between yard apps and the domain core.

Domain Core

- Core business rules for an APMC yard.
- **Rate Resolver:** Processes and labels rate provenance.
- **Auction & Weighing Logic:** Live bidding and recorded quantity processing.
- **Settlement Engine:** Server-computed bill, including weight-cut.

Data & External

- Persistent storage for all operational data.
- External data sources (like Agmarknet).



✓ Feasibility

A working prototype already exists – 200 source files, ~27,600 lines, 27 API routes, 447 seeded markets – and runs fully offline (cached Agmarknet snapshot, console-mode OTP), so it is demoable with zero live dependencies.

**200**

source files

**~27,600**

lines of code

**27**

API routes

**447**

seeded markets

**Works fully offline**

(cached Agmarknet snapshot, console-mode OTP)



⚠ Challenges & Risks



Patchy daily Agmarknet reporting across mandis



Low-end farmer handsets and connectivity



Adoption resistance from arhtiyas used to opaque pricing



🎯 Strategies



Provenance labels turn patchy data into an honest picture instead of a silent one.



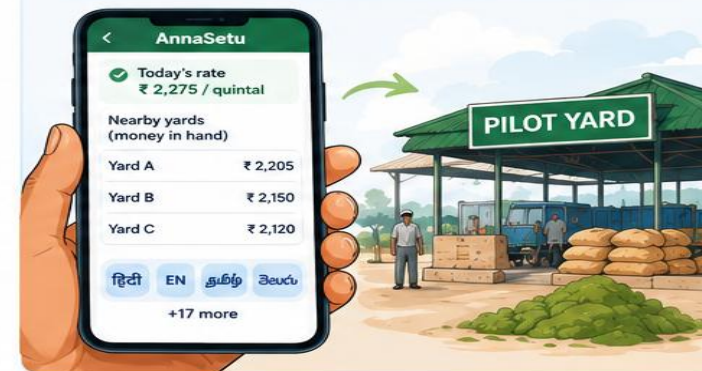
SMS + all-22-language support extends reach to feature phones.



AnnaSetu doesn't replace the auction or the agent, it only makes each existing step visible – which lowers adoption friction.



Rollout starts with a single pilot yard, as already demoed.



Impact on target audience:

- a farmer waiting at the yard sees a real wait-range instead of standing blind;
- a live auction & settlement status replaces uncertainty about procurement status;
- an itemised, server-computed bill (hamali only) ends silent deductions



Social:

- transparency and trust in APMC yards;
- farmer-readable information in all 22 official languages



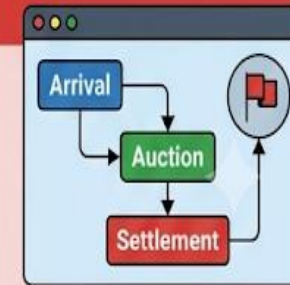
Economic:

- farmers compare nearby yards by money-in-hand after transport, not just headline rate, for better price realisation;
- corrected settlement math stops mis-charged commission



Governance:

- gives the Dept. a live, auditable trail of arrival → auction → settlement, with officer-only anomaly flags for oversight – nothing auto-penalised



- **Data source:** Agmarknet / data.gov.in – Directorate of Marketing & Inspection, Dept. of Agriculture & Farmers Welfare (real published rows, 15+ states, 12+ crops)
- **Domain input:** settlement formula verified in consultation with an APMC officer – hamali-only farmer deduction, commission and market fee buyer-borne
- **Live Prototype:** [Preview Here](#)
- **Source code:** [Github](#) – [README.md](#), [DATA-PROVENANCE.md](#), [AI-LAYER.md](#), [DECISIONS.md](#)